

WASP-36b

R-1893

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_160330 · created 2026-08-03T13:09:45+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457477.7114 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.048
Transit depth (Rp/Rs)^2	2.76 +/- 1.59 [%]
Orbital Inclination (inc)	82.26 +/- 2.89
Airmass coefficient 1 (a1)	1.16 +/- 0.73
Airmass coefficient 2 (a2)	0.02 +/- 0.26
Scatter in the residuals of the lightcurve fit is	63.48 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	6.75
Transit Duration (day)	0.064 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.048 vs 0.1368 (deviation 0.61σ)
Duration fit vs published	0.064 d vs 0.0758 d (deviation 0.41σ)
Mid-time O–C	4.0 min
Depth SNR	3.5
Residual scatter	63.48 %

Light curve

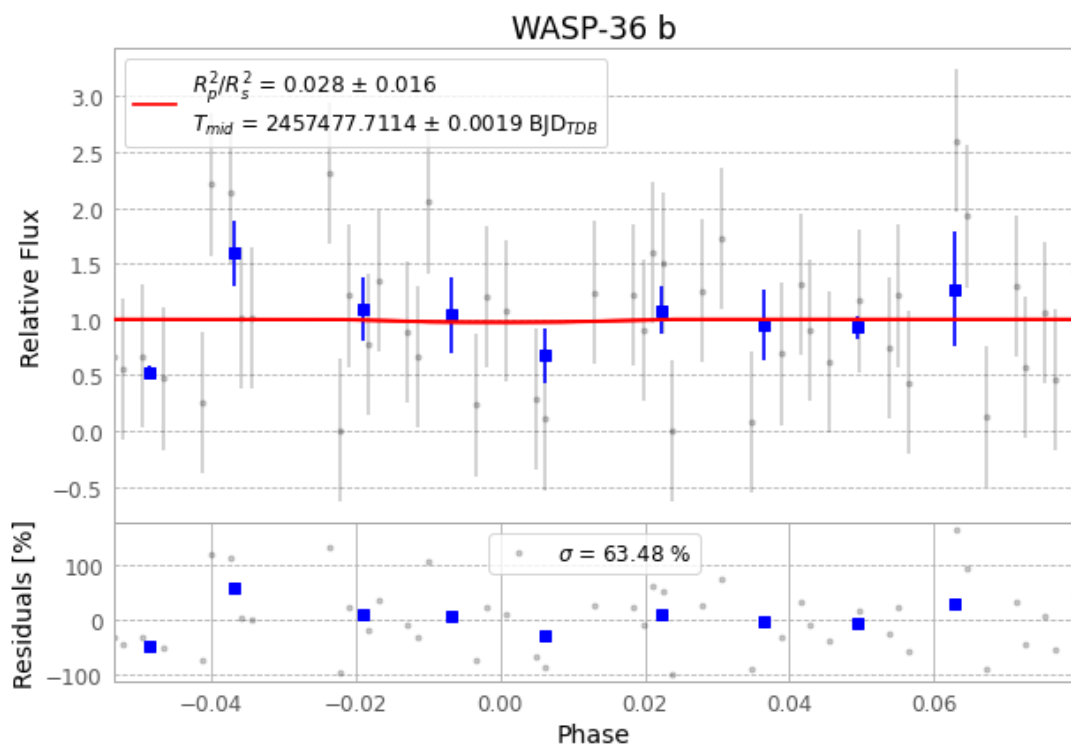


Figure 1 — FinalLightCurve_WASP-36 b_2016-03-29.png (SHA-256 305dbf9486a2e888...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [132, 160], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6fa980959620d1e9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-36160330025412.FITS → WASP-36160330075412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-160330.log (7478100b9b01...), FinalLightCurve_WASP-36 b_2016-03-29.png (305dbf9486a2...), FinalLightCurve_WASP-36 b_2016-03-29.csv (dc4d2f55d631...)

Compute resources

Wall time 265.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 13:09Z.